

COMPOSITE MATERIALS

Some definitions

A composite material is a combination of two or more chemically distinct and insoluble phases with recognizable interface, in such manner that its properties and structural performance are superior to those of the constituents acting independently.

Generally, one material forms a continuous matrix while the other provides the reinforcement.

Composites can be classified by their matrix material which include:

- Metal matrix composites (MMCs)
- Ceramic matrix composites (CMCs)
- Polymer matrix composites (PMCs) or sometimes referred to as organic matrix composites (OMCs)

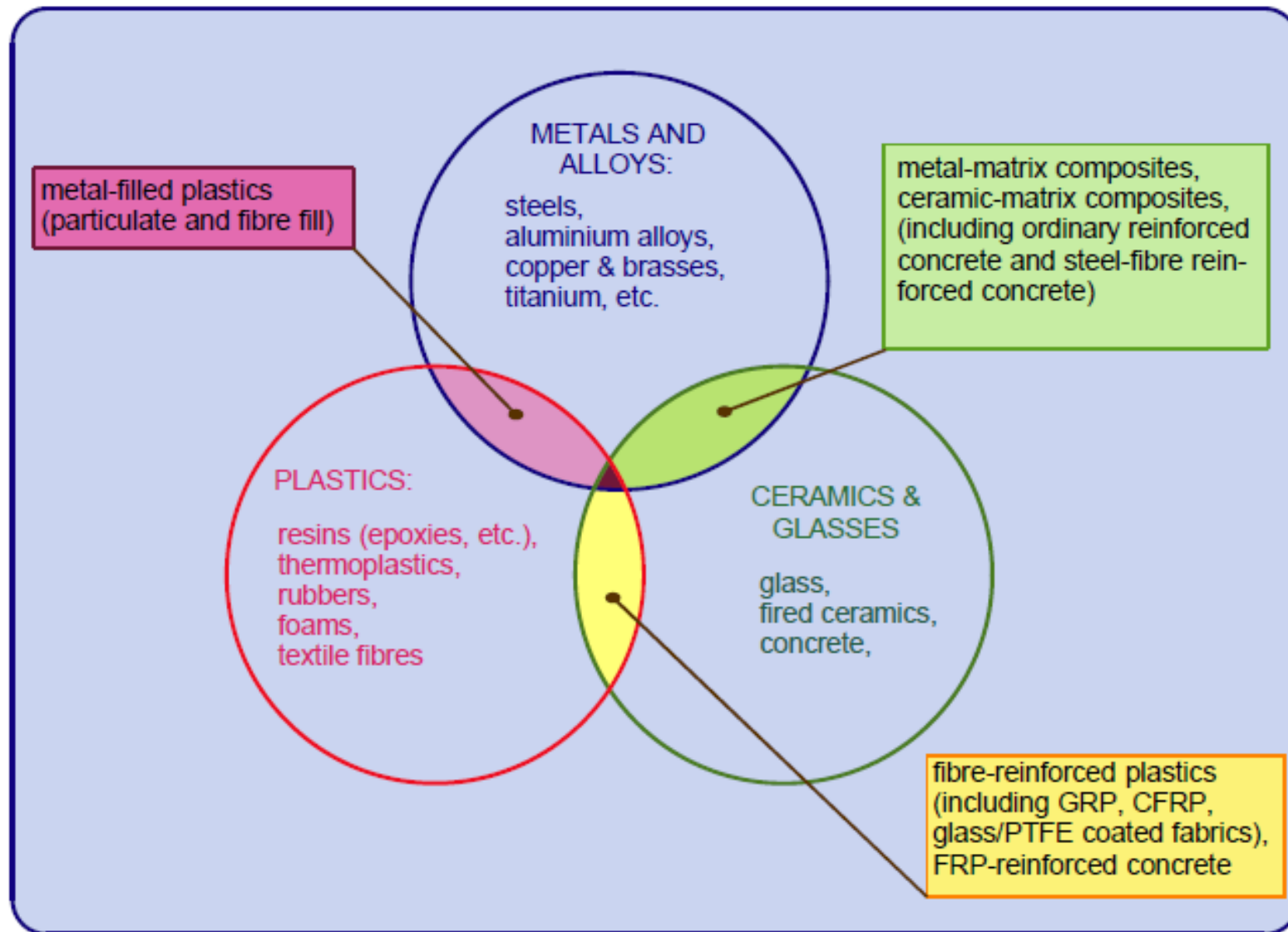


Figure 1.1. Relationships between classes of engineering materials, showing the evolution of composites

MMCs - Metal Matrix Composites

- The matrix is relatively soft and flexible.
- The reinforcement must have high strength and stiffness
- Since the load must be transferred from the matrix to the reinforcement, the reinforcement-matrix bond must be strong.

MMCs use:

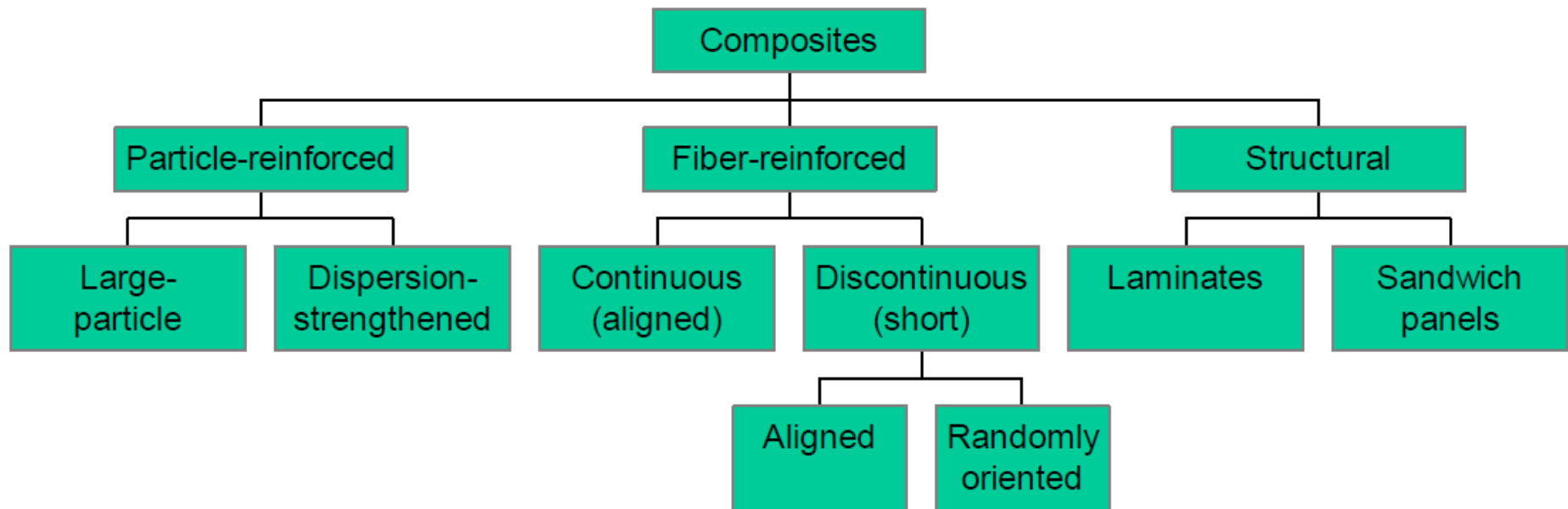
- Two types of particulates (dispersion strengthened alloys and regular particulate composites)
- Or long fiber reinforcements

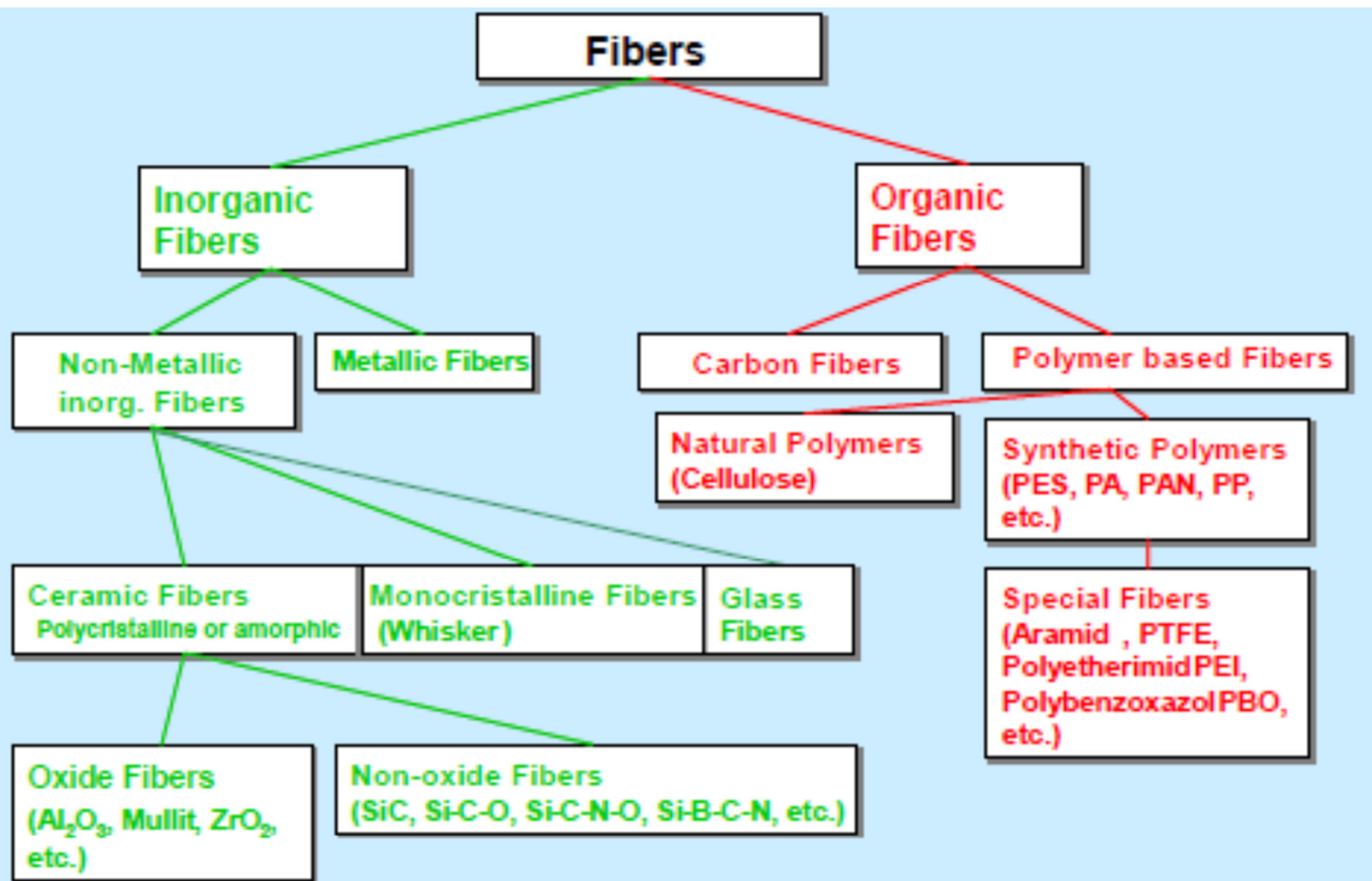
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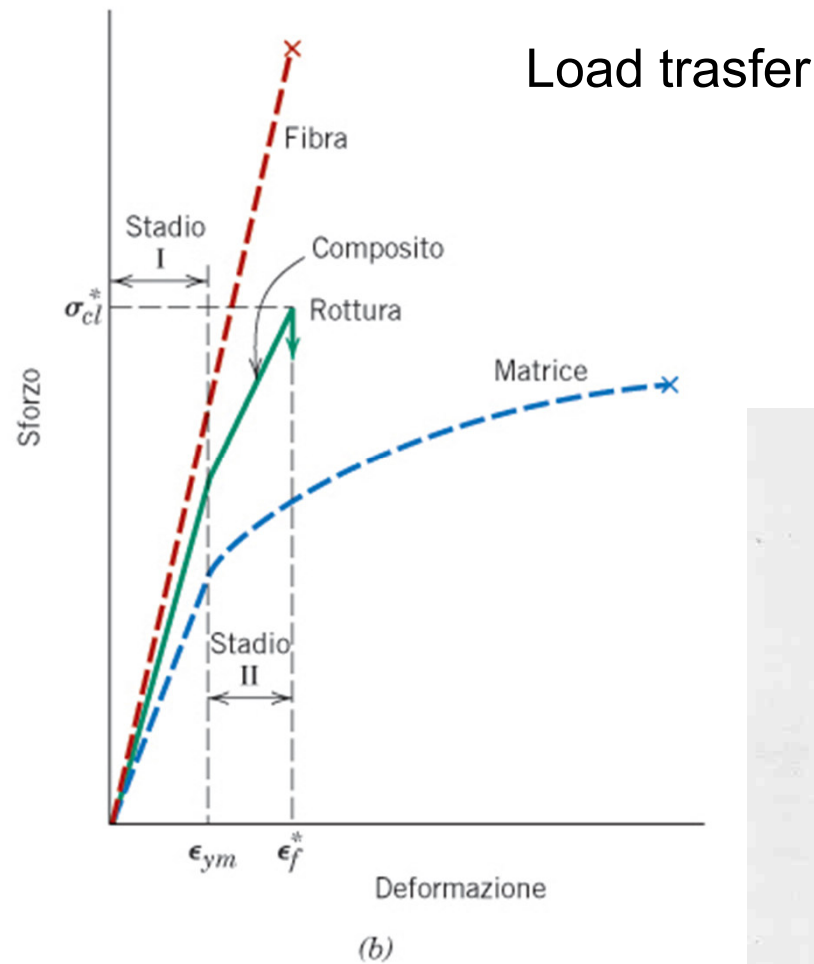
CMCs – Ceramic Matrix Composites

- The matrix is relatively hard and brittle
- The reinforcement must have high tensile strength to arrest crack growth
- The reinforcement must be free to pull out as a crack extends, so the reinforcement-matrix bond must be relatively weak

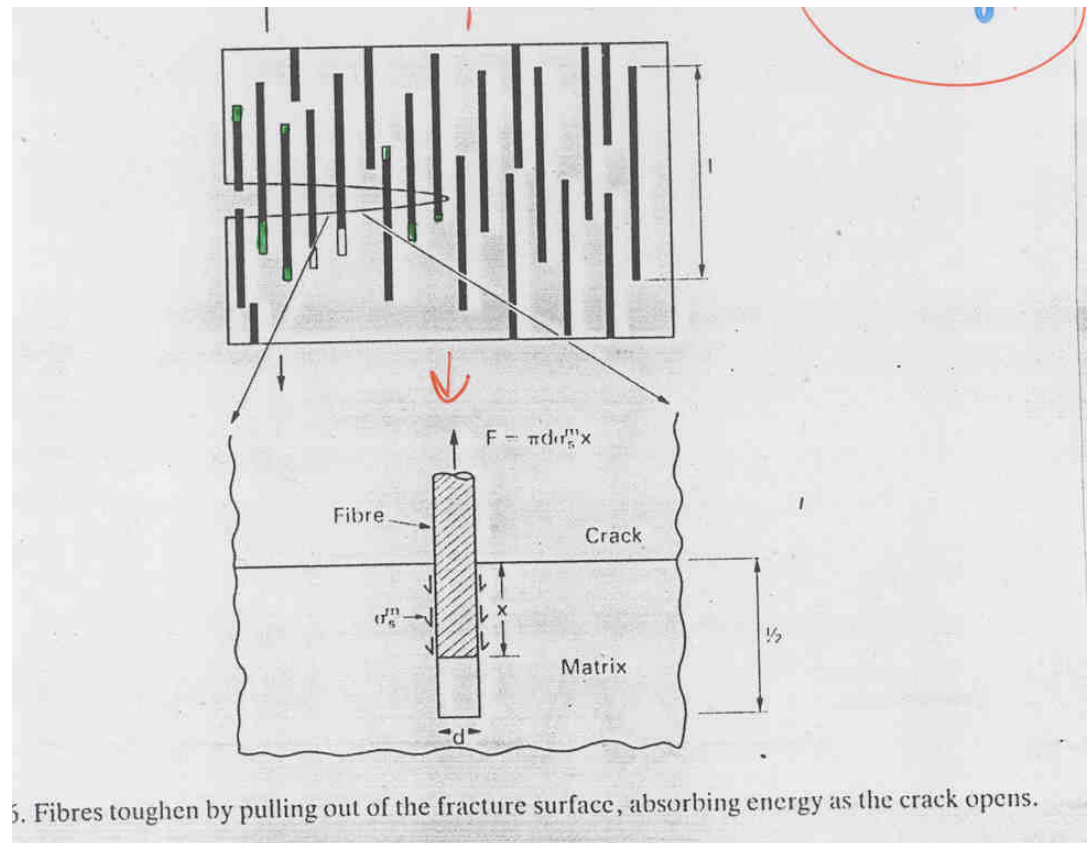




Reinforcement mechanisms due to fibres



Debonding (pull-out)



Materials Design: Strategy of Ceramic Matrix Composites

Polymer Matrix Composites PMC

- Fiber modulus $>$ Matrix modulus
- Fibers responsible for high strength and stiffness
- High fiber/matrix bondings
- No cracks
- Low porosity

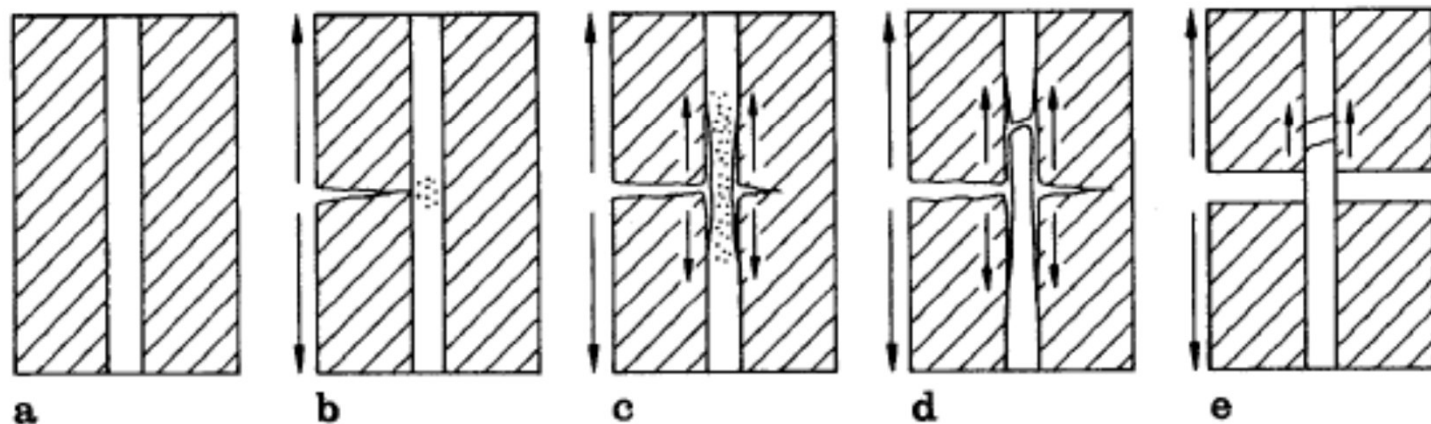
Monolithic Ceramics

- Material of high purity
- Small grain size
- Perfect grain boundaries
- No cracks, no defects
- No porosity

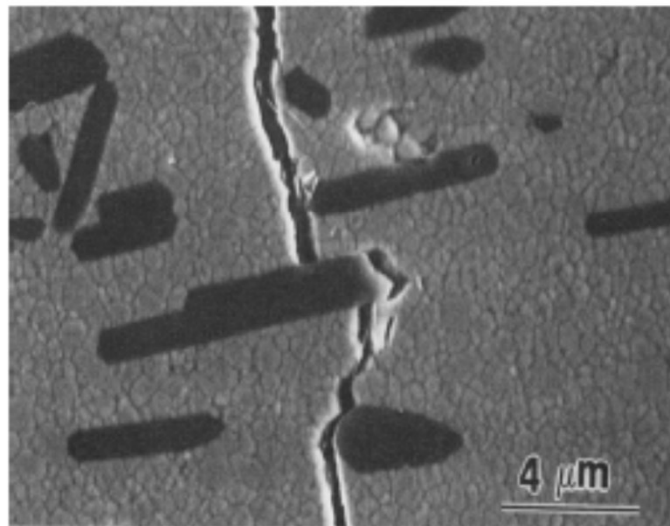
Ceramic Matrix Composites CMC

- Fiber modulus $\approx$ Matrix modulus
- Fibers lead to high damage tolerance
- Weak fiber/matrix bondings
- Microcracked matrix
- High porosity
- Heterogeneous microstructure

Toughening in ceramic-matrix composites



Schematic diagram showing mechanism of toughness gain in a brittle-matrix composite. (a) original state with frictional gripping, (b) matrix crack being temporarily halted by the fiber, (c) debonding and crack deflection along the fiber-matrix interface as a result of interfacial shear and lateral contraction of the fiber and matrix, (d) continued debonding, fiber failure at a weak point, and further crack extension, (e)



Crack deflection in a ceramic-ceramic (3Y-TZP- Al_2O_3) composite containing 15 vol% of 10–15 μm platelets. (*I. K. Cherian and W. M. Kriven, American Ceramics Society Bulletin, 80(12), December 2001, p. 57–67*)

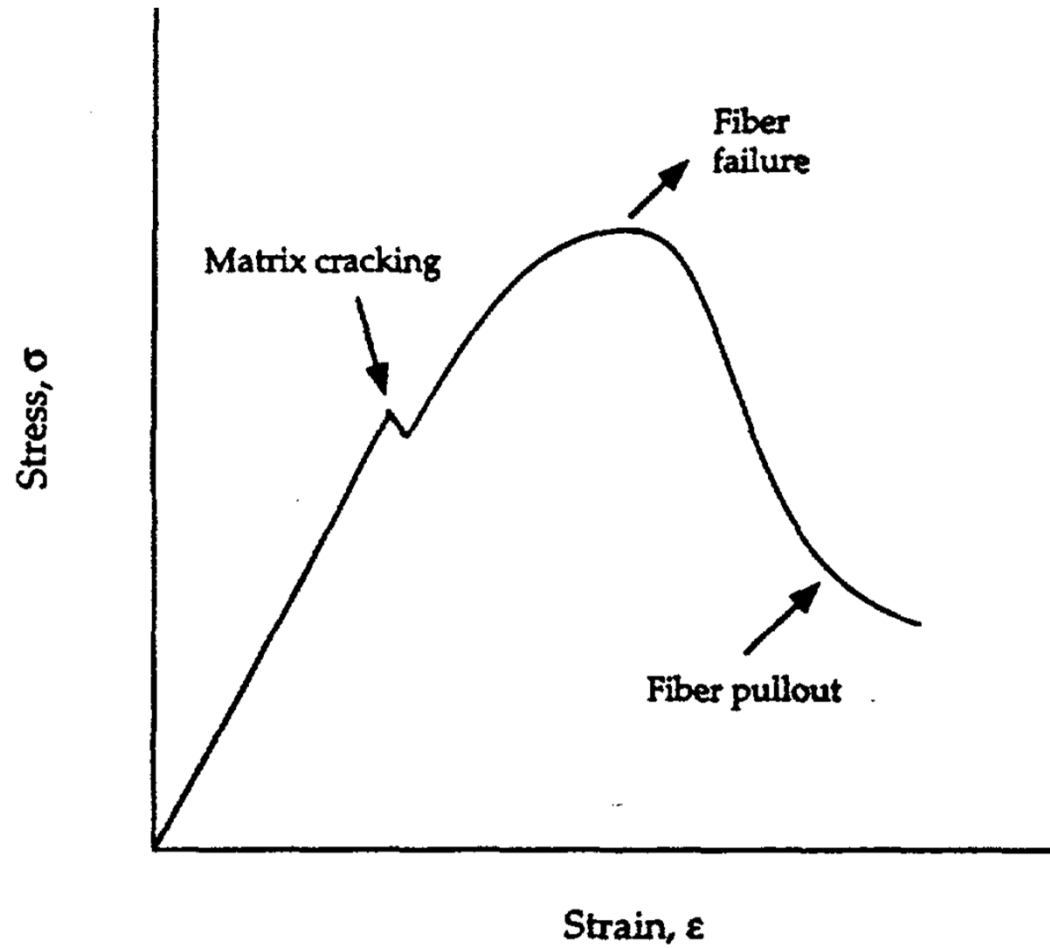
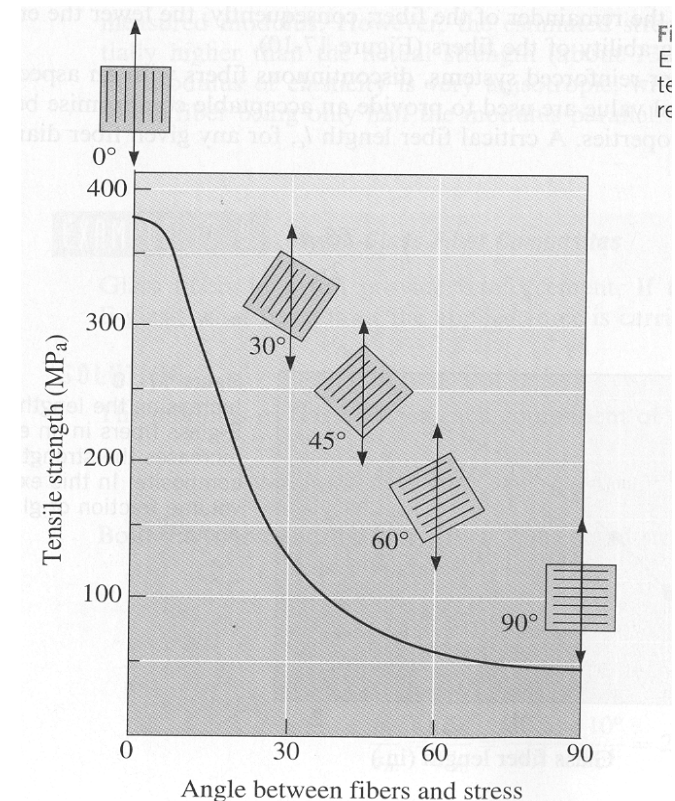
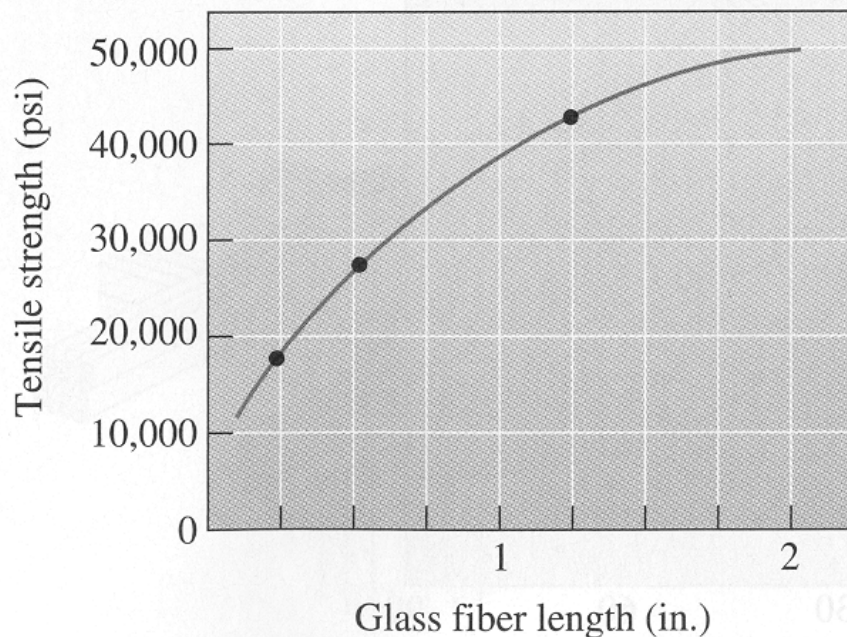
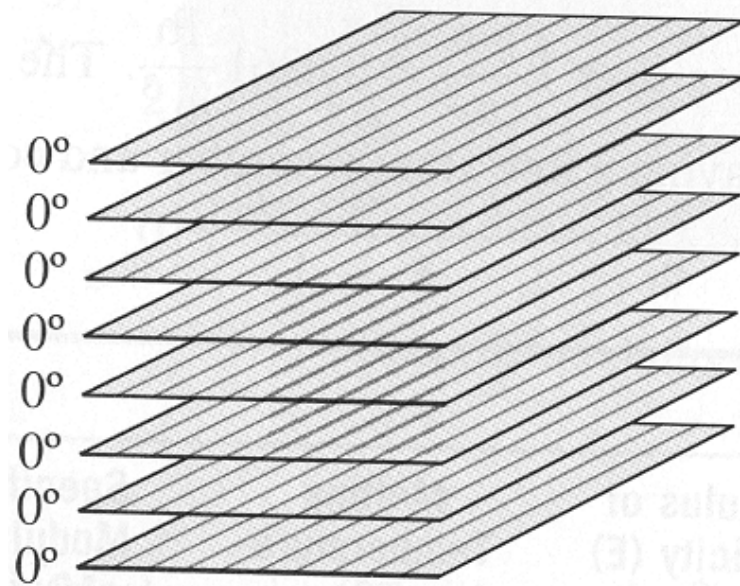


Fig. 2. A typical four-point flexure, stress-displacement curve for a CMC showing the various responses to increasing strain.

Fiber-reinforced polymer-matrix composites

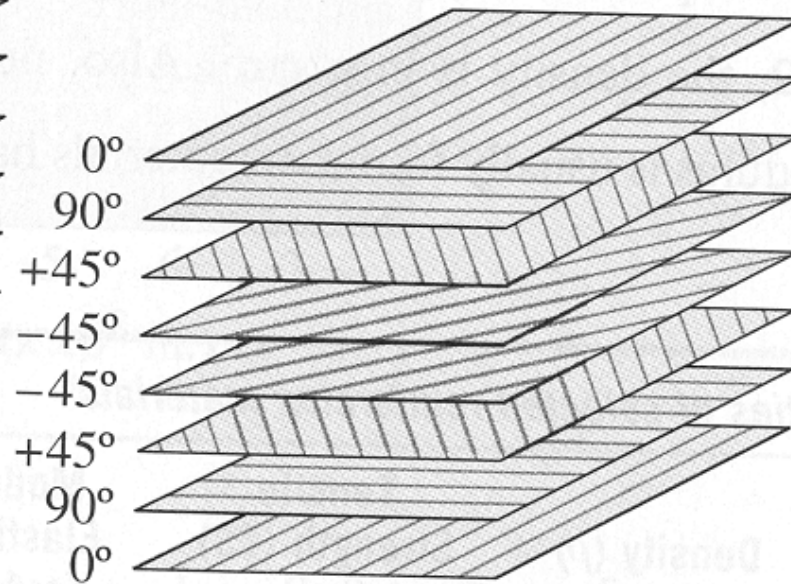
- Fiber reinforced composites provide improved strength, fatigue resistance, Young's modulus and strength to weight ratio over the constituent materials.
- This is achieved by incorporating strong, stiff, yet brittle fibers into a more ductile matrix.
- Generally speaking the fiber supplies the strength and stiffness while the matrix binds the fibers together and provides a means of transferring the load between fibers
- The matrix also provides protection for the fibers





Unidirectional

(a)



Cross-plyed
quasi-isotropic

(b)

Rule of mixtures

The rule of mixtures predicts some properties of the composite (e.g. density, elastic modulus, electrical conductivity, thermal conductivity) knowing the properties and volume fractions (f) of matrix and reinforcement.

$$\rho_c = f_m \rho_m + f_f \rho_f \quad \text{Note that } f_m = 1 - f_f$$

Where the subscripts m and f refer to the matrix and fiber.

The rule of mixtures can also be used to predict the modulus of elasticity when the fibers are continuous and unidirectional.

Parallel to the fibers, the modulus of elasticity may be as high as:

$$E_c = f_m E_m + f_f E_f$$

Perpendicular to the fibers, the modulus of elasticity may be as high as:

$$\frac{1}{E_c} = \frac{f_f}{E_f} + \frac{f_m}{E_m}$$

How is the mechanical advantage of composite measured?

$$\text{Specific modulus} = \frac{E}{\rho},$$

$$\text{Specific strength} = \frac{\sigma_{ult}}{\rho}.$$

where

E = Young's Modulus

ρ = Density

σ_{ult}

Ultimate Strength

**Specific modulus and strength of typical fibers,
composites and bulk metals**

Material	Specific Gravity	Young's Modulus	Ultimate Strength	Specific Modulus	Specific Strength
Units		GPa	MPa	GPa-m ³ /kg	MPa-m ³ /kg
Graphite	1.8	230.00	2067	0.1278	1.148
Unidirectional Graphite/Epoxy	1.6	181.00	1500	0.1131	0.9377
Cross-Ply Graphite/Epoxy	1.8	95.98	373.0	0.06000	0.2331
Quasi-isotropic Graphite/Epoxy	1.8	69.64	276.48	0.04353	0.1728
Steel	7.8	206.8	648.1	0.02652	0.08309
Aluminum	2.6	68.95	275.8	0.02652	0.1061

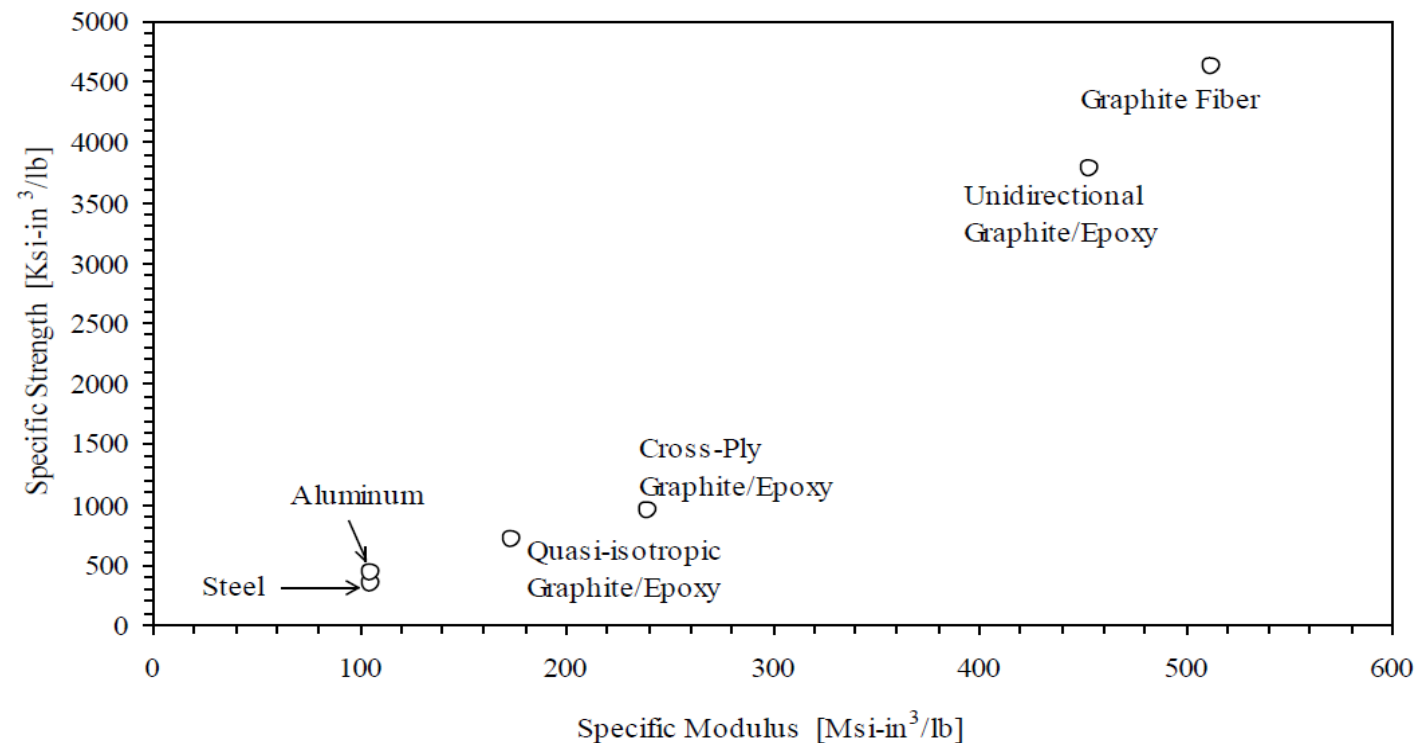
Some commonly used fibers for polymer matrix composites:

- Glass fibers
- Carbon fibers
- Aramid fibers

Some commonly used fibers for metal matrix composites:

- Boron fibers
- Carbon fibers
- Oxide ceramic and non-oxide ceramic fibers

Fibre	Cost (\$/lb)
Boron	320
Silicon carbide	100
Carbon	30
Alumina	30
Kevlar	20
E-glass	3

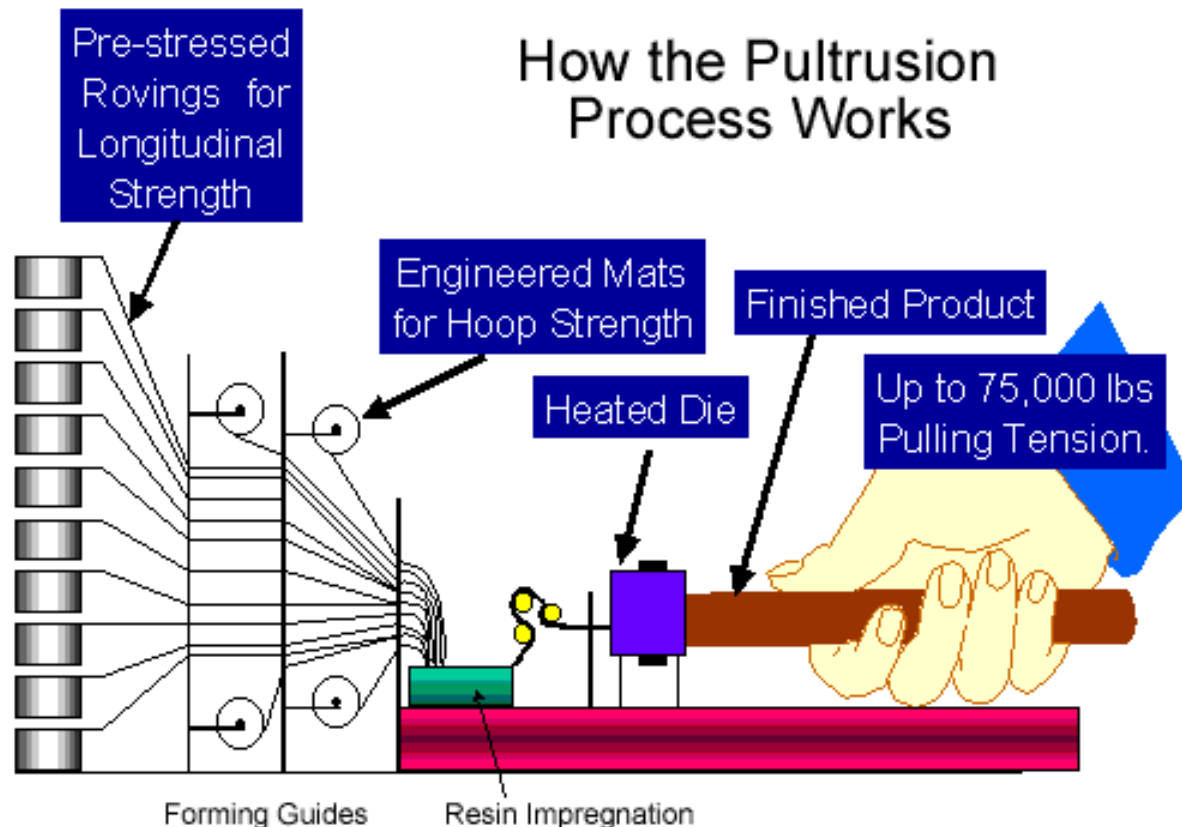


Carbon Fibre Reinforced Polymer

- Most common advanced composite
 - Carbon fibres are popular because
 - Highest specific modulus and specific strength of all reinforcing fibres
 - Retain stiffness and strength at high temperatures
 - Not affected by water, solvents or acids
 - Cost effective manufacturing process
 - Fibres are mixture of graphite and a-carbon
 - Uses
 - Sports equipment
 - Pressure vessels
 - Aircraft components
 - Anisotropic properties – woven or laminated
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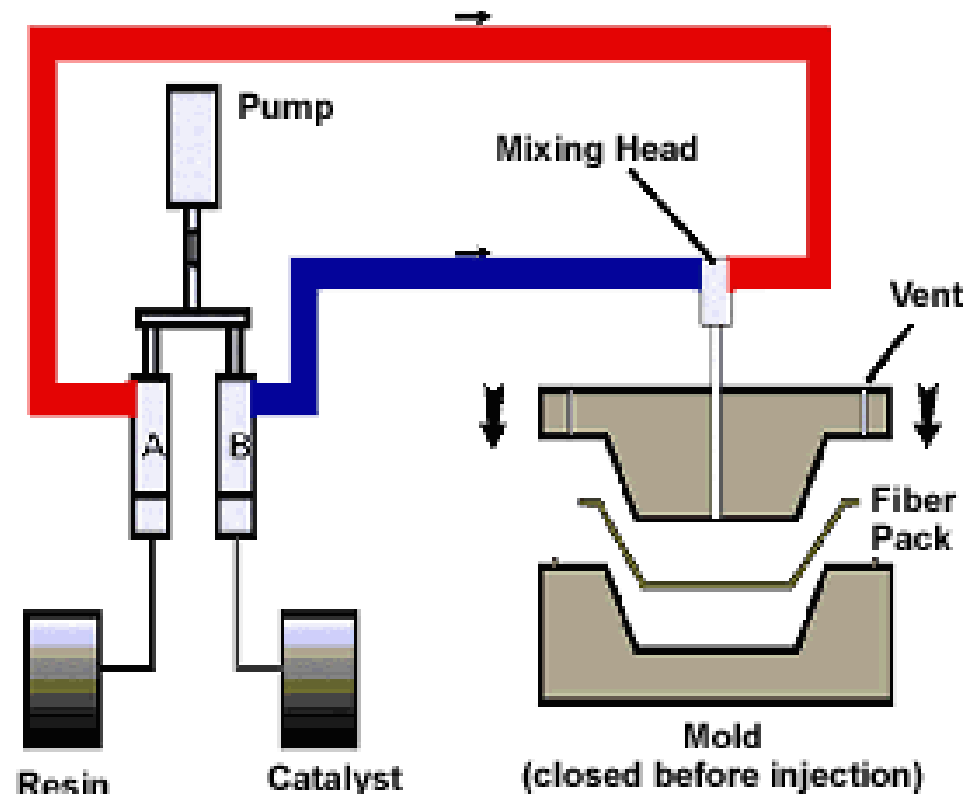
PULTRUSION

- Similar to extrusion of metallic parts
- Pultrusion involves pulling resin-impregnated glass strands through a die
- Standard extruded shapes can easily be produced such as pipes, channels, I-beams, etc.



RESIN TRANSFER MOLDING

- Resin transfer molding is a manufacturing method that is quite similar to injection molding where plastic is injected into a closed mold
- In the RTM process the preform (precut piece(s) of reinforcement) is placed in the mold, the mold is closed and the thermoset plastic matrix is injected into the mold, once the matrix is cured the part is ejected



Reinforcement mechanisms due to particles

